RIZ MUGHAL

QUIZ MASTER

Grand Quiz(PHY101)

100% correct solution.

For more information you can visit my channel and for any type of help related to CS619 you can contact me.

YOUTUBE CHANNEL:

https://www.youtube.com/channel/UCINsFwDiB62SValCcPDZbRQ/playlists

FACEBOOK GROUP:

https://www.facebook.com/groups/923887914750307

PHY101:Grand Quiz

Question # 1 of 30 (Start time: 10:35:12 AM, 04 January 2021)

The ultimate strength of a sample is the stress at which the sample:

Select the correct option

○	returns to its original shape when the stress is removed
○	remains underwater
◉	breaks
○	bends 180?

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PHY101:Grand Quiz

Question # 2 of 30 (Start time: 10:35:36 AM, 04 January 2021

The value of k in coulomb's law depends upon

Select the correct option

- (●) medium between two charges
- ( ) distance between charges
- ( ) magnitude of charges
- ( ) all of these

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PHY101:Grand Quiz

Question # 3 of 30 (Start time: 10:35:54 AM, 04 January 202

The law of inertia was firstly formulated by:

Select the correct option

- ○ Aristotle
- ○ Galileo
- ◉ Newton
- ○ Einstein

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Question # 4 of 30 (Start time: 10:36:10 AM, 04 January 2021)

The ratio of circumference of a circle to its diameter is equal to:

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Select the correct option

- ○ 2 π
- ◉ π
- ○ π/2
- ○ one steradian

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Quiz Start Time: 10:35 AM, 04 January 2021

Question # 5 of 30 (Start time: 10:36:25 AM, 04 January 2021) **Total Marks:** 1

A water bed that is 1.5 m wide and 2.5 m long weighs 1055 N. Assuming the entire lower surface of the bed is in contact with the floor, what is the pressure the bed exerts on the floor?

Select the correct option

- ○ 250 Pa
- ○ 260 Pa
- ○ 270 Pa
- ◉ 280 Pa

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PHY101:Grand Quiz

Question # 6 of 30 (Start time: 10:36:40 AM, 04 January 2021)

The centre of gravity is the average location of the ..... of an object

Select the correct option

- mass
- weight
- static equilibrium
- dynamic

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Question # 7 of 30 (Start time: 10:36:54 AM, 04 January 2021)

Acceleration in a body is always produced in the direction of:

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Select the correct option

- ○ Velocity
- ○ Weight
- ◉ Force
- ○ Acceleration

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Quiz Start Time: 10:35 AM, 04 Jar

Question # 8 of 30 (Start time: 10:37:10 AM, 04 January 2021) Tot

As per Coulomb's law, the force of attraction or repulsion between two point charges directly proportional to the

Select the correct option

- ○ cube of the distance
- ◉ product of the magnitude of charges
- ○ sum of the magnitude of charge
- ○ square of the distance between them

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PHY101:Grand Quiz G

Question # 9 of 30 (Start time: 10:37:25 AM, 04 January 2021)

One revolution is the same as:

Select the correct option

- [x] 2π rad
- [ ] π rad
- [ ] 57 rad
- [ ] 1 rad

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Question # 10 of 30 (Start time: 10:37:39 AM, 04 January 2021)

A ---------- vector is obtained by dividing the vector by its magnitude:

Select the correct option

- (●) unit
- ( ) position
- ( ) normal
- ( ) negative

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Question # 11 of 30 (Start time: 10:37:57 AM, 04 January 2021) **Total Marks:**

Take the speed of sound to be 340m/s. A thunder clap is heard about 3 s after the lightning is seen. The source of both light and sound is:

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Select the correct option

- ○ moving overhead faster than the speed of sound
- ○ emitting a much higher frequency than is heard
- ○ emitting a much lower frequency than is heard
- ◉ about 1000m away

PHY101:Grand Quiz

Quiz Start Time: 10:35 AM, 04 Januar

Question # 12 of 30 (Start time: 10:38:14 AM, 04 January 2021) **Total M**

If the distance between all pairs of particles of the body do not change by applying a force then the body is said to be

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Select the correct option

- (●) rigid
- ( ) large
- ( ) small
- ( ) flexible

PHY101:Grand Quiz

Quiz Start Time: 10:35 AM, 04 January 2021

Question # 13 of 30 (Start time: 10:38:34 AM, 04 January 2021) **Total Marks:** 1

For an ideal fluid flowing through a horizontal pipe, Bernoulli's equation states that the sum of the pressure and energy per unit volume along the pipe does which of the following? (Assume measurements are taken along the pipe in the direction of fluid flow.)

Select the correct option

- ○ increases as the pipe diameter increases
- ○ decreases as the pipe diameter increases
- ● remains constant as the pipe diameter increases
- ○ increases, then decreases as the pipe diam[illegible]eases

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Question # 14 of 30 (Start time: 10:38:59 AM, 04 January 2021)

Which of the following statement is true?

Select the correct option

- (●) Weight is a force, mass is a measure of inertia
- ( ) Mass depends on gravity, weight does not
- ( ) Gravity is necessary to measure both weight and mass
- ( ) Heavier objects weigh more than light objects

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PHY101:Grand Quiz

Quiz Start Time: 10:35 AM, 04 Januar

Question # 15 of 30 (Start time: 10:39:14 AM, 04 January 2021)

Total M

A/an ---------- is the basic reason to change in the motion of an object according to Newton's second law of motion.

Select the correct option

- (●) net force
- ( ) decrease in inertia
- ( ) change in velocity
- ( ) acceleration

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Quiz Start Time: 1

Question # 16 of 30 (Start time: 10:39:30 AM, 04 January 2021)

When a spring is compressed or stretched, the potential energy of the spring

Select the correct option

- ○ Decreases
- ○ Stays constant
- ◉ Increases
- ○ Becomes zero

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Question # 17 of 30 (Start time: 10:39:44 AM, 04 January 2021)

The dimensional units of ratio of work and power is:

Select the correct option

- J
- T
- L
- F

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Question # 18 of 30 (Start time: 10:39:58 AM, 04 January 2021)

A particle oscillating in simple harmonic motion is:

Select the correct option

○	in equilibrium at the ends of its path because its velocity is zero there in equilibrium at the ends of its path because its velocity is zero there
◉	in equilibrium at the center of its path because the acceleration is zero there
○	never in equilibrium because it is in motion
○	never in equilibrium because there is always a force

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Quiz Start Time: 10:35 AM, 04 January 2021

Question # 19 of 30 (Start time: 10:40:13 AM, 04 January 2021) **Total Marks:**

A fire whistle emits a tone of 170 Hz. Take the speed of sound in air to be 340m/s. The wavelength of this sound is about:

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Select the correct option

- ○ 3.0m
- ○ 0.5m
- ○ 1.0m
- ◉ 2.0m

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Quiz Start Time: 10:35 AM, 04 January 2021

Question # 20 of 30 (Start time: 10:40:28 AM, 04 January 2021) **Total Marks:** 1

How much pressure is exerted on a submarine at a depth of 8.50 km in the Pacific Ocean? (The density of sea water = 1.025 * 103 kg/m3, and the atmospheric pressure at sea level = 1.01 * 105 Pa.)

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Select the correct option

- ○ 8.6 * 105 Pa
- ○ 8.7 * 106 Pa
- ○ 9.5 * 106 Pa
- ◉ 8.6 * 106 Pa

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Quiz Start Time:

Question # 21 of 30 (Start time: 10:40:44 AM, 04 January 2021)

A force of 100 N acts upon a body for five seconds. What will be the change in momentum?

Select the correct option

- ○ 200 NS
- ◉ 500 NS
- ○ 20 NS
- ○ 50 NS

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Question # 22 of 30 (Start time: 10:40:59 AM, 04 January 2021)

One newton is a force that produces an acceleration of 0.5 m/s2 in a body of mass:

Select the correct option

- 1 kg
- 2 kg
- 4 kg
- 8 kg

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Quiz Start

Question # 23 of 30 (Start time: 10:41:15 AM, 04 January 2021)

When breaks are applied to a fast moving car, the passenger will be throws:

Select the correct option

- (●) Forward
- ( ) Backward
- ( ) Downward
- ( ) Upward

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Question # 24 of 30 (Start time: 10:41:30 AM, 04 January 2021)

In the formula F= Gm1m2/r2, the quantity G:

Select the correct option

○	depends on the local value of g
○	is used only when Earth is one of the two masses
○	is greatest at the surface of Earth
◉	is a universal constant of nature

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Question # 25 of 30 (Start time: 10:41:45 AM, 04 January 2021)

Per-second refers to the dimensions of physical quantity:

Select the correct option

- ○ Angular displacement
- ◉ Angular velocity
- ○ Angular acceleration
- ○ Angular momentum

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Question # 26 of 30 (Start time: 10:41:59 AM, 04 January 2021)

Light year is a unit of:

Select the correct option

- acceleration
- velociy
- time
- distance

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Question # 27 of 30 (Start time: 10:42:18 AM, 04 January 2021)

Young's modulus can be used to calculate the strain for a stress that is:

Select the correct option

- ○ just below the ultimate strength
- ○ just above the ultimate strength
- ◉ well below the yield strength
- ○ well above the yield strength

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PHY101:Grand Quiz Quiz Start Tim

Question # 28 of 30 (Start time: 10:42:32 AM, 04 January 2021)

A wheel of radius 50 cm having the angular speed of 5 rad/s will have linear speed in m/s?

Select the correct option

- ○ 0.5
- ○ 1.5
- ◉ 2.5
- ○ 3.5

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Quiz Start Time: 10:35 AM, 0

Question # 29 of 30 (Start time: 10:42:47 AM, 04 January 2021)

An object moves in a circle at constant speed. The work done by the centripetal force is zero because:

Select the correct option

○	the displacement for each revolution is zero
○	the average force for each revolution is zero
○	there is no friction
◉	the centripetal force is perpendicular to the velocity

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Quiz Start Time: 10:35 AM, 04 January 2021

Question # 30 of 30 (Start time: 10:43:04 AM, 04 January 2021) **Total Marks:** 1

A ball is thrown upward into the air with a speed that is greater than terminal speed. It lands at the place where it was thrown. During its flight the force of air resistance is the greatest:

Select the correct option

- (●) just after it is thrown
- ( ) half way up
- ( ) at the top of its trajectory
- ( ) halfway down

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